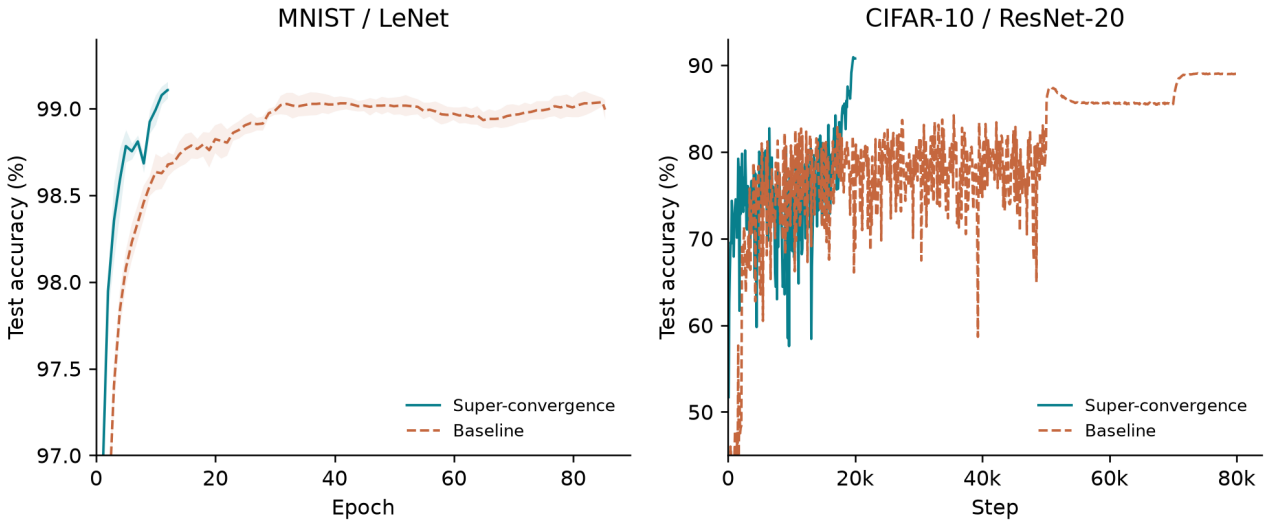
# Super-Convergence: reproduction

**Hardware:** Apple M3 Pro, 36 GB RAM. PyTorch 2.2.2 / MPS, float32.

**MNIST:** LeNet, batch 512, three seeds. 1cycle: LR 0.01 → 0.1 → 0.00001; inverse-decay baseline.
**CIFAR-10:** ResNet-20, batch 1,000, one seed. CLR: LR 0.1 → 3 → 0.1; step-decay baseline.

| Experiment | Schedule | Paper (%) | My result (%) | Steps |
|---|---|---|---|---|
| MNIST | Baseline | 99.03 ± 0.04 | 98.99 ± 0.06 | 10,000 |
| | 1cycle | 99.25 ± 0.03 | 99.11 ± 0.05 | 1,406 |
| CIFAR-10 | Baseline | 88.60 | 88.99 | 80,000 |
| | CLR | 90.40 | 90.77 | 20,000 |




Compared with the baseline, super-convergence trained 7.3× faster on MNIST and 4.0× faster on CIFAR-10. CIFAR-10 training took 1.34 hours with CLR and 5.37 hours with the baseline, excluding test evaluation.

The accuracy gain on CIFAR-10 was +1.78 points; the paper reports +1.80. On MNIST, 1cycle fell 0.14 points below the paper.

Partial reproduction: PyTorch instead of Caffe; CIFAR replicas simulated on one GPU. One CIFAR seed is not enough to assess run-to-run variation.